Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A bar magnet of magnetic moment $4.0\ \mathrm{A}-\mathrm{m}^2$ is free to rotate about a vertical axis through its center. The magnet is released from rest from eastwest position. Kinetic energy of the magnet in north-south position will be $(\mathrm{H}=25 \mu \mathrm{T})$

(A) $10^{-2}\ \mathrm{J}$

(B) $10^{-4}\ \mathrm{J}$

(C) $10^{-6}\ \mathrm{J}$

(D) 0

Q.2

A short bar magnet of magnetic moment $0.4\ \mathrm{J}\ \mathrm{T}^{-1}$ is placed in a uniform magnetic field of $0.16\ \mathrm{T}$. The magnet is in stable equilibrium when the potential energy is

(A) $-0.64\ \mathrm{J}$

(B) zero

(C) $-0.082\ \mathrm{J}$

(D) $-0.064\ \mathrm{J}$

Q.3

A coil in the shape of an equilateral triangle of side l is suspended between the pole pieces of a permanent magnet such that B is in plane of the coil. If due to a current i in the triangle a torque τ acts on it, the side l of the triangle is

(A) $\frac{2}{\sqrt{3}}\left(\frac{\tau}{\mathrm{B.i}}\right)^{\frac{1}{2}}$

(B) $2\left(\frac{\tau}{\sqrt{3}\,\mathrm{Bi}}\right)^{\frac{1}{2}}$

(C) $\frac{2}{\sqrt{3}}\left(\frac{\tau}{\mathrm{Bi}}\right)$

(D) $\frac{1}{\sqrt{3}}\frac{\tau}{\mathrm{B.i}}$

Q.4

A bar magnet of magnetic moment $3.0\ \mathrm{A}-\mathrm{m}^2$ is placed in a uniform magnetic field of $2 \times 10^{-5}\ \mathrm{T}$. If each pole of the magnet experiences a force of $6 \times 10^{-4}\ \mathrm{N}$, the length of the magnet is

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(A) $0.5\ \mathrm{m}$

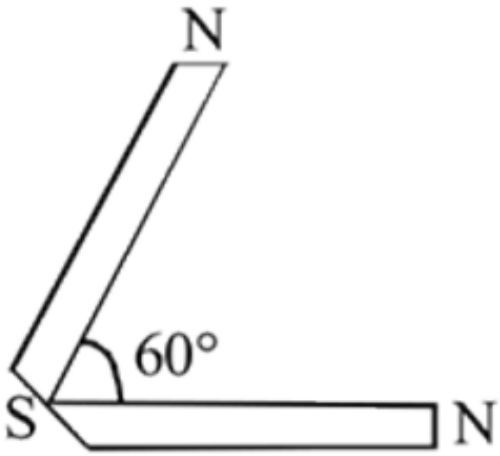
(B) $0.3\ \mathrm{m}$

(C) $0.2\ \mathrm{m}$

(D) $0.1\ \mathrm{m}$

Q.5

The net magnetic moment of two identical magnets each of magnetic moment M_0, inclined at 60° with each other is



(A) M_0

(B) $\sqrt{2}\mathrm{M}_0$

(C) $\sqrt{3}\mathrm{M}_0$

(D) $2\mathrm{M}_0$

Q.6

Two identical magnetic dipoles of magnetic moments $1.0\ \mathrm{A}-\mathrm{m}^2$ each, placed at a separation of $2\ \mathrm{m}$ with their axis perpendicular to each other. The resultant magnetic field at a point midway between the dipoles is

(A) $5 \times 10^{-7} T$

(B) $\sqrt{5} \times 10^{-7} T$

(C) $10^{-7}\ \mathrm{T}$

(D) None of these

Q.7

A magnetic needle is kept in a non-uniform magnetic field. It experiences

(A) neither a force nor a torque

(B) a torque but not a force

(C) a force but not a torque

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(D) a force and a torque

Q.8

Force between two identical bar magnets whose centres are r metre apart is 4.8 N, when their axes are in the same line. If separation is increased to $2r$, the force between them is

(A) 2.4 N

(B) 1.2 N

(C) 0.6 N

(D) 0.3 N

Q.9

A magnetic needle lying parallel to a magnetic field requires W units of work to turn it through $60°$. The torque required to maintain the needle in this position will be

(A) $\sqrt{3}W$

(B) W

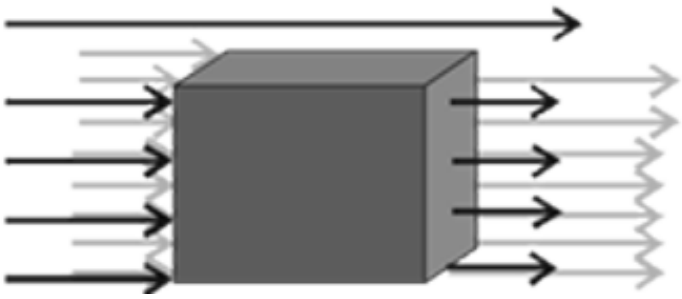
(C) $\frac{\sqrt{3}}{2}W$

(D) $2W$

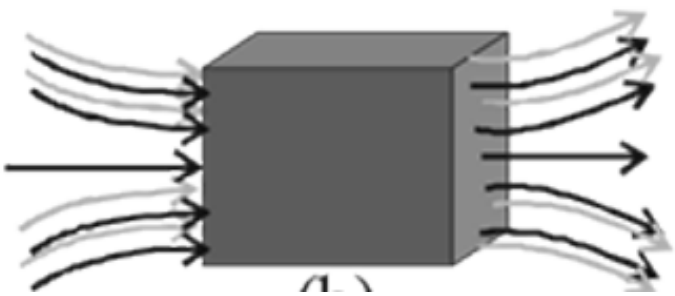
Q.10

A uniform magnetic field parallel to the plane of the paper exists in space initially directed from left to right. When a bar of soft iron is placed in the field parallel to it, the lines of force passing through it will be represented by

(A)



(B)

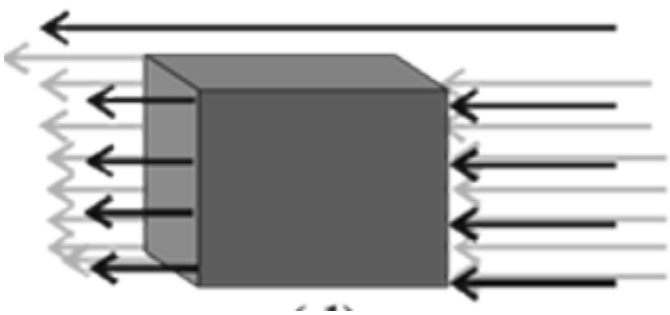


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(C)



(D)



Q.11

A bar magnet having centre O has a length of 4 cm. Point P_1 is in the broad side-on and P_2 is in the end side-on position with $\mathrm{OP}_1 = \mathrm{OP}_2 = 10$ metres. The ratio of magnetic intensities H at P_1 and P_2 is

(A) $\mathrm{H}_1 : \mathrm{H}_2 = 16 : 100$

(B) $\mathrm{H}_1 : \mathrm{H}_2 = 1 : 2$

(C) $\mathrm{H}_1 : \mathrm{H}_2 = 2 : 1$

(D) $\mathrm{H}_1 : \mathrm{H}_2 = 100 : 16$

Q.12

A bar magnet $8\mathrm{cms}$ long is placed in the magnetic merdian with the N-pole pointing towards geographical north. Two netural points separated by a distance of $6\mathrm{cms}$ are obtained on the equatorial axis of the magnet. If horizontal component of earth's field $= 3.2 \times 10^{-5}\ \mathrm{T}$, then pole strength of magnet is

(A) $5\mathrm{ab} - \mathrm{amp} \times \mathrm{cm}$

(B) $10\mathrm{ab} - \mathrm{amp} \times \mathrm{cm}$

(C) $2.5\mathrm{ab} - \mathrm{amp} \times \mathrm{cm}$

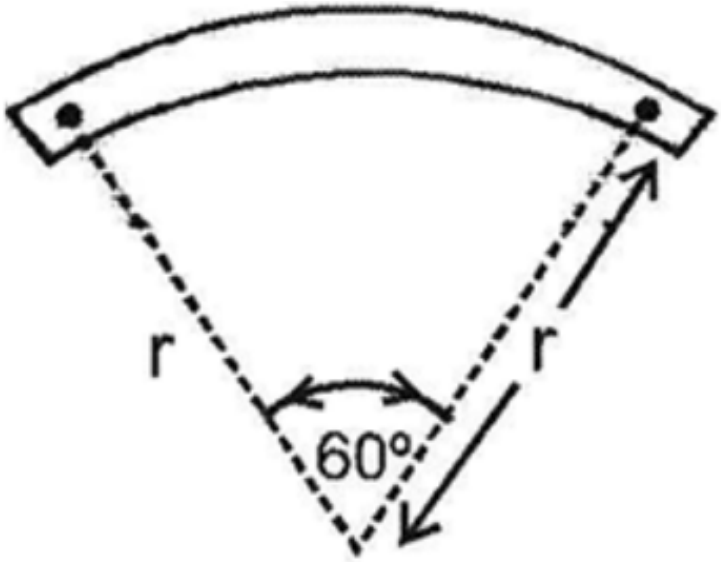
(D) $20\mathrm{ab} - \mathrm{amp} \times \mathrm{cm}$

Q.13

A bar magnet of length ' ℓ ' and magnetic dipole moment 'M' is bent in the form of an arc as shown in figure.

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The new magnetic dipole moment will be



(A) $\frac{3}{\pi}\mathrm{M}$

(B) $\frac{2}{\pi}\mathrm{M}$

(C) $\frac{M}{n}$

(D) M

Q.14

Two isolated point poles of strength $30\ \mathrm{A}-\mathrm{m}$ and $60\ \mathrm{A}$ -m are placed at a distance of $0.3\ \mathrm{m}$. The force of repulsion is

(A) $2 \times 10^{-3}\ \mathrm{N}$

(B) $2 \times 10^{-4}\ \mathrm{N}$

(C) $2 \times 10^{5}\ \mathrm{N}$

(D) $2 \times 10^{-5}\ \mathrm{N}$

Q.15

A magnet of magnetic moment $50\hat{i}\ \mathrm{A}-\mathrm{m}^2$ is placed along the x -axis in a magnetic field $\vec{B}=(0.5\hat{i}+3.0\hat{j})T$. The torque acting on the magnet is

(A) $175\hat{k}\ \mathrm{N}-\mathrm{m}$

(B) $150\hat{k}\ \mathrm{N}-\mathrm{m}$

(C) $75\hat{k}\ \mathrm{N}-\mathrm{m}$

(D) $25\sqrt{37}\hat{k}\ \mathrm{N}-\mathrm{m}$

Q.16

A magnetic dipole is under the influence of two magnetic fields. The angle between the field directions is 60°

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and one of the fields has a magnitude of 1.2×10^{-2} T. If the dipole comes to stable equilibrium at an angle of 15° with this field, what is the magnitude of other field?

(A) 4.4×10^{-3} tesla

(B) 5.2×10^{-3} tesla

(C) 3.4×10^{-3} tesla

(D) 7.8×10^{-3} tesla

Q.17

Two identical short bar magnets, each baving magnetic moment M, are placed at distance of $2d$ apart with axis perpendicular to each other in a horizontal plane. The magnetic induction at a point midway between them is

(A) $\frac{\mu_0}{4\pi}(\sqrt{2})\frac{M}{d^3}$

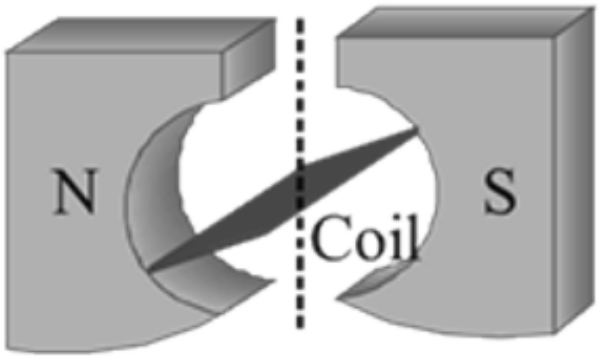
(B) $\frac{\mu_0}{4\pi}(\sqrt{3})\frac{M}{d^3}$

(C) $\left(\frac{2\mu_0}{\pi}\right)\frac{M}{d^3}$

(D) $\frac{\mu_0}{4\pi}(\sqrt{5})\frac{M}{d^3}$

Q.18

The figure below shows the north and south poles of a permanent magnet in which n turn coil of area of cross-section A is resting, such that for a current i passed through the coil, the plane of the coil makes an angle θ with respect to the direction of magnetic field B. If the plane of the magnetic field and the coil are horizontal and vertical respectively, the torque on the coil will be



(A) $\tau = niAB\cos\theta$

(B) $\tau = niAB\sin\theta$

(C) $\tau = niAB$

(D) None of the above, since the magnetic field is radial

Q.19

A magnetic dipole is acted upon by two magnetic fields which are inclined to each other at an angle of 75°.

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One of the fields has a magnitude of 15 mT. The dipole attains stable equilibrium at an angle of 30° with this field. The magnitude of the other field (in mT) is close to :

(A) 1

(B) 11

(C) 36

(D) 1060

Q.20

A vibration magnetometer consists of two identical bar magnets placed one over the other such that they are mutually perpendicular and bisect each other. The time period of the combination in a magnetic field is 4 s. If one of the magnets is removed, then the time period of another one in the same field will be

(A) 5.78 s

(B) 3.36 s

(C) 4.36 s

(D) 5.36 s

Answers & Solutions

Q.1 Answer:

10^{-4} J

Solution:

Loss in P.E. = gain in K.E. $\therefore E_k = U_i - U_i = -MB\cos 90^{\circ} - (-MB\cos 0^{\circ})$
$= 4 \times 25 \times 10^{\frac{1}{6}} = 10^{-4}$ J

Q.2 Answer:

-0.064 J

Solution:

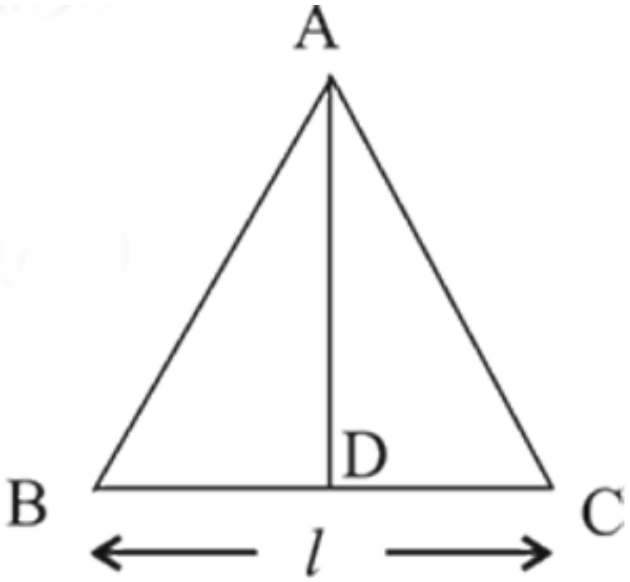
For stable equilibrium $U = -MB = -(0.4)(0.16) = -0.064$ J

Q.3 Answer:

$2\left(\frac{\tau}{\sqrt{3}\,Bi}\right)^{\frac{1}{2}}$

Solution:

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$= \frac{1}{2}(l)\sqrt{l^2 - \left(\frac{l}{2}\right)^2} = \frac{\sqrt{3}}{4}l^2$

$\Rightarrow \frac{\sqrt{3}}{4}(l)^2 = \frac{\tau}{\mathrm{Bi}} \Rightarrow \therefore l = 2\left(\frac{\tau}{\sqrt{3}\,\mathrm{Bi}}\right)^{\frac{1}{2}}$

Q.4 Answer:

$0.1\ \mathrm{m}$

Solution:

$F = mB$ **or** $6 \times 10^{-4} = m \times 2 \times 10^{-5}$

$\therefore \quad m = 30\ \mathrm{A-m}$

Now, $\ell = \frac{M}{m} = \frac{3}{30} = 0.1\ \mathrm{m}$

Q.5 Answer:

$\sqrt{3}\mathrm{M}_0$

Solution:

$$\mathrm{M}_{\text{net}} = \sqrt{\mathrm{M}_0^2 + \mathrm{M}_0^2 + 2\mathrm{M}_0^2 \cos 60^\circ}$$

$$= \sqrt{3\mathrm{M}_0^2} = \sqrt{3}\mathrm{M}_0$$

Q.6 Answer:

$\sqrt{5} \times 10^{-7} T$

Solution:

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$$B_1 = \frac{\mu_0}{4\pi} \cdot \frac{2M}{r^3} = 10^{-7} \times \frac{2 \times 1}{1^3} = 2 \times 10^{-7} T$$

$$\text{and } B_2 = \frac{\mu_0}{4\pi} \cdot \frac{M}{r^3} = 10^{-7} \times \frac{1}{1^3} = 10^{-7} T$$

$$\text{Now, } B = \sqrt{B_1^2 + B_2^2} = \sqrt{5} \times 10^{-7} T.$$

Q.7 Answer:

a force and a torque

Solution:

A magnetic needle kept in non uniform magnetic field experience a force and torque due to unequal forces acting on poles

Q.8 Answer:

0.3 N

Solution:

Force between two short bar magnets is given by $F = \frac{\mu_0}{4\pi} \frac{6M_1M_2}{r^4}$

$\therefore \frac{F_1}{F_2} = \frac{r_2^4}{r_1^4} = \left(\frac{2r}{r}\right)^4$

or $F_2 = \frac{F_1}{16} = \frac{4.8}{16} = 0.3 \text{ N}.$

Q.9 Answer:

$\sqrt{3}W$

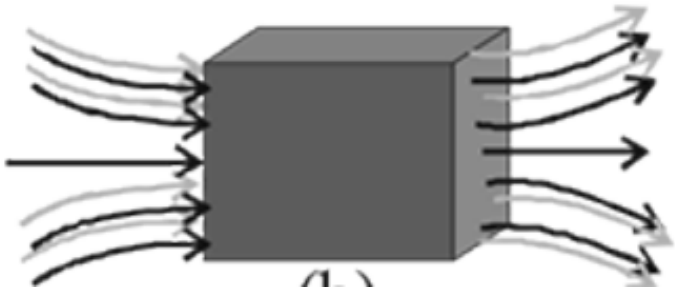
Solution:

$W = MB \cos 60^\circ$

and $\tau = MB \sin 60^\circ$

$\therefore \tau = \sqrt{3}W.$

Q.10 Answer:



Solution:

Permeability of soft iorn is maximum, so maximum lines of force tries to pass through the soft iron.

Q.11 Answer:

$H_1 : H_2 = 1 : 2$

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Solution:

Magnetic intensity on end side -on position is twice than broad side on position

Q.12 Answer:

$5\text{ab} - \text{amp} \times \text{cm}$

Solution:

Here, $2\ell = 8\text{ cm}, \ell = 4\text{ cm}, \ \text{d} = \frac{6}{2} = 3\text{ cm}.$

At neutral point, $\text{H} = \text{B} = \frac{\mu_0}{4\pi}\frac{\text{M}}{(\text{d}^2+\ell^2)^{3/2}}$

$= 10^{-7}\frac{\text{M}}{(5\times10^{-2})^3} = \frac{\text{M}}{1250}$

$\therefore \quad \text{M} = 1250\text{H} = 1250 \times 3.2 \times 10^{-5}\text{Am}^2$

$\text{m} = \frac{\text{M}}{2\ell} = \frac{1250\times3.2\times10^{-5}}{8\times10^{-2}}\ \text{A m.} = 0.5$

$\text{Am} = 0.5 \times \frac{1}{10}\text{abamp} \times 100\text{ cm}$

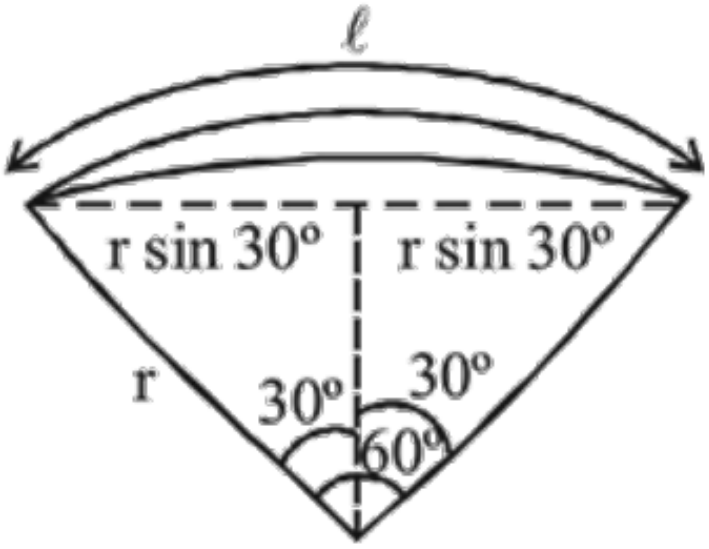
$= 5\text{ab} - \text{amp} \times \text{cm}$

Q.13 Answer:

$\frac{3}{\pi}\text{M}$

Solution:

Magnetic dipole moment $\text{M} = \text{m} \times \ell \quad \text{M}' = \text{m} \times \text{r}$



Q.14 Answer:

$2 \times 10^{-3}\text{ N}$

Solution:

$\text{F} = \frac{\mu_0}{4\pi}\frac{\text{m}_1\,\text{m}_2}{\text{r}^2} = 10^{-7} \times \frac{30\times60}{(0.3)^2} = 2 \times 10^{-3}\text{ N}$

Q.15 Answer:

$150\hat{k}\text{ N} - \text{m}$

Solution:

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$\vec{\tau} = \vec{M} \times \vec{B} = 50\hat{i} \times (0.5\hat{i} + 3\hat{j}) = 150\hat{k}N - m$

Q.16 Answer:

4.4×10^{-3} tesla

Solution:

Given that : $\mathrm{B}_1 = 1.2 \times 10^{-2}$ T, **orientation of dipole with the field** B_1, $\theta_1 = 15^\circ$ **Hence, orientation of dipole with** B_2, $\theta_2 = 60^\circ - 15^\circ = 45^\circ$ **(figure)**

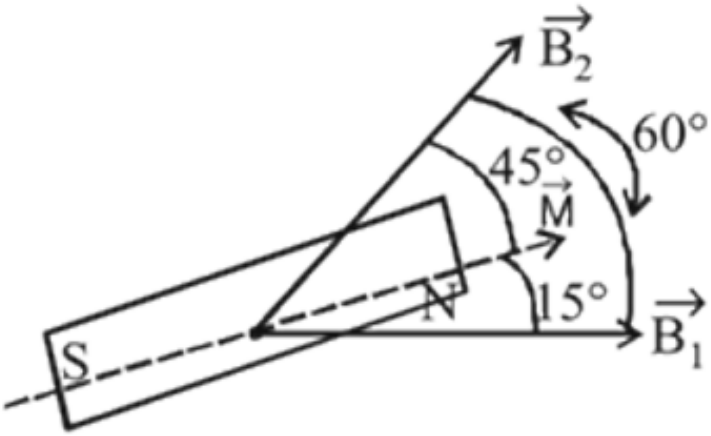
As the dipole is in equilibrium, therefore, the torque on the dipole due to the two fields must be equal and opposite. If M **be the magnetic dipole moment of the dipole, then**

$\tau_1 = \tau_2$ **or**

$\mathrm{MB}_1 \sin\theta_1 = \mathrm{MB}_2 \sin\theta_2$

or, $B_2 = \frac{B_1 \sin\theta_1}{\sin\theta_2} = \frac{1.2\times10^{-2}\sin 15^\circ}{\sin 45^\circ}$

$= \frac{1.2\times10^{-2}\times0.2588}{0.7071} = 4.4 \times 10^{-3}$ **Tesla**



Q.17 Answer:

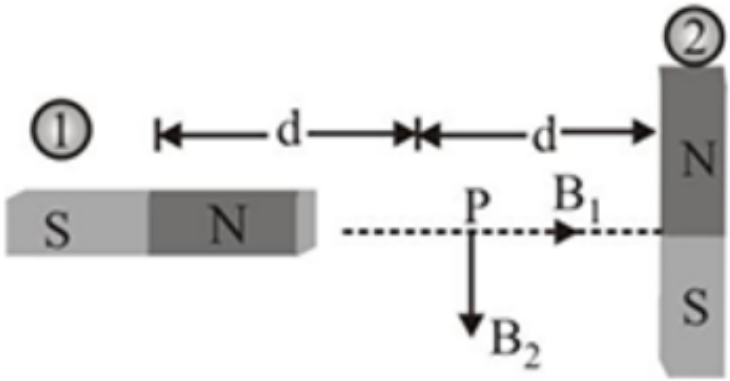
$\frac{\mu_0}{4\pi}(\sqrt{5})\frac{M}{d^3}$

Solution:

At point P **net magnetic field** $B_{\text{set}} = \sqrt{B_1^2 + B_2^2}$

where $B1 = \frac{\mu_0}{4\pi} \cdot \frac{2M}{d^3}$ **and** $B_2 = \frac{\mu_0}{4\pi} \cdot \frac{M}{d^3}$

$\Rightarrow B_{mt} = \frac{\mu_0}{4\pi} \cdot \frac{\sqrt{5}MA}{d^3}$



Q.18 Answer:

$\tau = niAB\cos\theta$

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Solution:

Plane of coil is having angle θ with the magnetic field. $\therefore \tau = MB\sin(90^\circ - \theta)$ **or** $\tau = niAB\cos\theta$(**As** M $= \mathrm{niA}$)

Q.19 Answer:

11

Solution:

We know that, magnetic dipole moment $\mathrm{M} = \mathrm{NiA}\cos\theta$ **i.e.,** $\mathrm{M} \propto \cos\theta$

When two magnetic fields are inclined at an angle of 75° **the equilibrium will be at** 30°**, so** $\cos\theta = \cos(75^\circ - 30^\circ) = \cos 45^\circ = \frac{1}{\sqrt{2}}$

$\frac{x}{\sqrt{2}} = \frac{15}{2} \therefore x \approx 11$

Q.20 Answer:

3 .36 s

Solution:

As magnets are perpendicular to each other, the resultant magnetic moment

$$\mathrm{M'} = \sqrt{\mathrm{M}^2 + \mathrm{M}^2} = \sqrt{2}\ \mathrm{M}$$

$$\therefore \quad \mathrm{T}_1 = 2\pi\sqrt{\frac{2\mathrm{I}}{\sqrt{2}\,\mathrm{MB}}}$$

In the second case, $\mathrm{T}_2 = 2\pi\sqrt{\frac{\mathrm{I}}{\mathrm{MB}}}$

$$\frac{\mathrm{T}_2}{\mathrm{T}_1} = \frac{1}{2^{1/4}}$$

$$\therefore \quad \mathrm{T}_2 = \frac{4}{2^{1/4}} = 3\ .36\ \mathrm{s}$$

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